

Advanced (Habanero)

Q1. Solve the system: $x + y = 3$, $x^2 + y^2 = 9$

- (A) $(x, y) = (1, 2)$
(B) $(x, y) = (2, 1)$
(C) $(x, y) = (0, 3)$
(D) $(x, y) = (3, 0)$
(E) No solution

Q2. Find the intersection point of $y = 2x - 1$ and $y = x^2 - 4$

- (A) $(-1, -3)$
(B) $(2, 3)$
(C) $(1, 1)$
(D) $(-2, -5)$
(E) $(3, 5)$

Q3. Solve the system: $y = x + 1$, $y = x^2 - 1$

- (A) $x = 0, y = 1$
(B) $x = 1, y = 2$
(C) $x = -1, y = 0$
(D) $x = 2, y = 3$
(E) No solution

Q4. Find the solution to the system: $x - y = 2$, $x^2 + y^2 = 8$

- (A) $(2, 0)$
(B) $(0, -2)$
(C) $(-2, 0)$
(D) $(0, 2)$
(E) $(1, -1)$

Q5. Solve for x and y : $y = 2x + 1$, $y = x^2 + 1$

- (A) $x = 1, y = 3$
(B) $x = -1, y = -1$
(C) $x = 0, y = 1$
(D) $x = 2, y = 5$
(E) $x = -2, y = -3$

Q6. Find the intersection points of $y = x^2 - 2x + 1$ and $y = 2x - 1$

- (A) $(1, 1)$
(B) $(0, -1)$
(C) $(2, 3)$
(D) $(-1, -3)$
(E) $(1, 0)$

Q7. Solve the system of equations: $y = x^2 + 1$, $y = x + 2$

- (A) $x = -1, y = 2$
(B) $x = 1, y = 3$
(C) $x = 0, y = 1$
(D) $x = 2, y = 5$
(E) No solution

Q8. Find the solution to $x + y = 4$, $x^2 + y^2 = 16$

- (A) $(4, 0)$
(B) $(0, 4)$
(C) $(2, 2)$
(D) $(-2, -2)$
(E) $(-4, 0)$

Open Ended Questions (FRQ)**Q9.** Solve the system: $y = 2x - 1$, $y = x^2 - 2x - 3$. Show all work and find the intersection points.**Q10.** Find the solution to the system: $x - 2y = 3$, $x^2 + 2y^2 = 11$. Use algebraic methods and show all work.**Chef's Tips**

- Emphasize the importance of checking solutions in the original equations.
- Encourage students to use graphing to visualize the intersection points.
- Remind students to provide clear and concise work for the free-response questions.